ChatGPT

CA TUCR HTLA Continuous to Discrete Bridge Formalization

Executive assessment

Your July 18 test log already established the important architectural fact: the continuous layer can drive a scalar sparsity gate that performs a k-WTA wipeout on the 1024D quinary substrate, while the masking rule enforces the invariant $P_A = 0 \Rightarrow P_S = 0$ even under an injected shock. The same log also made the remaining gaps explicit: no ratified dC_r/dt, no ratified 666D-to-threshold reduction, and no ratified nontrivial $W_C, W_E, W_R, W_N, \Lambda$. The recommendations below preserve the documented structure and replace only those three open placeholders with bounded, executable mathematics. filecite turn0file0

The proposal keeps the existing ODE skeleton and the existing k-WTA bridge, but makes the bridge mathematically tighter in three ways. First, cognitive resources become a bounded leaky integrator slaved to an explicit energy-budget variable, which makes the 85% cap exact rather than aspirational. Second, the 666D coupling contribution becomes a spectral friction term driven by the dominant real growth mode of the coupling matrix, with a conservative non-normality guard. Third, the 11D coupling matrices are instantiated as normalized neurocomputational gains consistent with the best-supported directional motifs in the literature: reward-prediction-error strongly drives dopamine; novelty drives norepinephrine and acetylcholine; fast error drive is routed through glutamatergic excitation; GABA supplies distributed inhibition; serotonin contributes a net stabilizing and suppressive influence on runaway excitation and internal gain. [1]

Resource dynamics

A resource law for this architecture has to do two jobs at once. It has to act like a slow allostatic reserve, because brains regulate behavior under finite energetic constraints, and it has to crash hard when surprise and state-velocity spike, because those are exactly the conditions that should tighten the Lyapunov fail-safe. The neuroscience side of that choice is well motivated: signaling is metabolically expensive, the brain operates under a finite energy budget, and predictive/allostatic regulation is fundamentally about matching energetic supply to anticipated demand. [2]

I recommend introducing one auxiliary scalar $B(t)$, the **remaining admissible expendable budget**, with $0 \le B(t) \le 0.85 E_{\max}$, and then defining $C_r(t)$ as a filtered accessibility of that budget rather than the budget itself. Let

$$p(t) = \alpha_C \, \|\dot{C}(t)\|_2^2 + \alpha_R \, RPE(t)^2 + \alpha_N \, Novelty(t)^2$$

with normalized gains

$$\alpha_C = 0.25, \qquad \alpha_R = 0.18, \qquad \alpha_N = 0.14.$$

Then evolve the budget by

$$\dot{B}(t) = -\frac{p(t)}{0.85E_{\max}}\,B(t), \qquad B(0) = 0.85E_{\max}.$$

Now define the resource state directly as

$$\dot{C}_r(t) = \rho\left(0.05 + 0.95\frac{B(t)}{0.85E_{\max}} - C_r(t)\right) - p(t)\,\big(C_r(t) - 0.05\big),$$

with

$$\rho = 0.08, \qquad C_r(t) \in [0.05, 1].$$

This is the ratified dC_r/dt. It is a true leaky integrator because the first term relaxes C_r toward the budget-limited ceiling $0.05 + 0.95\,B/(0.85E_{\max})$, while the second term depletes accessible resources proportionally to instantaneous energetic demand. Because $p(t)$ depends on both $\|\dot{C}\|_2^2$ and the squared shock variables, the law naturally crashes under large RPE and novelty spikes, which is exactly the behavior the failure protocol wants. [3]

Two invariants come for free. If $C_r = 0.05$, then

$$\dot{C}_r = \rho\left(0.95\frac{B}{0.85E_{\max}}\right) \geq 0,$$

so the floor is forward invariant. If $C_r = 1$, then the relaxation term is nonpositive and the depletion term is strictly nonpositive, so $\dot{C}_r \leq 0$, which makes the ceiling forward invariant. Even more importantly, the 85% budget cap is exact:

$$\int_0^T \Big(-\dot{B}(t)\Big)\,dt = B(0) - B(T) \leq 0.85E_{\max}.$$

So the architecture can never admit more than $0.85E_{\max}$ of expendable bridge energy over the episode encoded by B. That is the cleanest way to satisfy the conservation requirement without changing the rest of the bridge. [4]

Spectral friction law

For the 666D coupling term, the right object is not an RMS norm. In continuous-time systems, instability is governed by the dominant real growth mode. For normal systems, the sign of the largest real eigenvalue determines asymptotic growth or decay; in neural-network stability theory, nonpositive real parts are the key absolute-stability condition. In reservoir-style recurrent systems, spectral dominance is also the standard proxy for when recurrent activity is approaching unstable amplification. [5]

Let the 666D coupling object be a real square matrix $M \in \mathbb{R}^{666\times 666}$. Define the spectral-abscissa term

$$\alpha(M) = \max_i \operatorname{Re}\lambda_i(M).$$

Because highly non-normal matrices can transiently amplify even when $\alpha(M) < 0$, I recommend one conservative guard term based on the symmetric part,

$$\mu_2(M) = \lambda_{\max}\left(\frac{M + M^\top}{2}\right),$$

and then the effective resonance score

$$\widehat{\alpha}(M) = \max\{\alpha(M), \mu_2(M)\}.$$

This is still fully spectral, and it does exactly what you want operationally: it reacts both to a truly unstable dominant eigenmode and to matrices whose symmetric part can produce large Euclidean growth before asymptotic decay settles in. That is a robustness guard, not a redesign. [6]

The scalar friction penalty should then be

$$\Pi_M = \gamma_M \left(\exp\left(\beta_M [\widehat{\alpha}(M)]_+\right) - 1\right), \qquad [x]_+ = \max(x, 0),$$

with

$$\gamma_M = 28, \qquad \beta_M = 2.4.$$

This choice is deliberate. When the dominant mode is stable, $\widehat{\alpha}(M) \le 0$, the friction vanishes. Once the dominant mode crosses into positive growth, the penalty rises exponentially rather than linearly, which is the mathematically appropriate way to "violently tighten" the gate against runaway resonance. For reference, $\widehat{\alpha} = 0.5$ yields a penalty of about 65 active coordinates, while $\widehat{\alpha} = 1.0$ yields about 281, which nearly chokes the legal 307-coordinate active ceiling by itself.

Preserving your documented 11D affine gate form, the finalized threshold becomes

$$k_{11D}(t) = k_0 + a\,DA(t) - b\,5HT(t) - c\,GABA(t) - d\,Temp(t),$$

with the normalized coefficients

$$k_0 = 256, \qquad a = 96, \qquad b = 64, \qquad c = 72, \qquad d = 32,$$

and

$$k_{\text{allow}}(t) = \left\lfloor \operatorname{clip}\left(k_{11D}(t) - \Pi_M,\ 1,\ 307\right) \right\rfloor.$$

I am not changing the documented gate structure from your empirical log; I am only making the 666D reduction exact and giving the affine 11D term a bounded calibration compatible with the new invariant cube $C \in [0,1]^{11}$. The fail-safe remains as documented: if $\|\dot{C}\|_2 > 5C_r$, halve k_{allow} again after the spectral penalty has already been applied. filecite turn0file0

Cross-coupling matrices

The 11-state order is

$$[NE,\ DA,\ 5HT,\ ACh,\ Glu,\ GABA,\ Perception,\ Cognition,\ Affect,\ Volition,\ Self]^\top.$$

The guiding biological motifs are straightforward and well supported. Dopamine is the canonical reward-prediction-error teaching signal and should therefore be the strongest direct recipient of RPE input. Novelty and unexpected uncertainty recruit norepinephrine strongly and acetylcholine strongly. Glutamate is the principal fast excitatory transmitter and should carry external error drive into perceptual and cognitive state variables. GABA should supply broad inhibitory balancing pressure. Serotonin is not purely inhibitory across all receptor classes, but the strongest architecture-level abstraction for your stabilizing channel is a net suppressive influence on spontaneous gain, excitatory throughput, and affective volatility, with a modest positive contribution to self-stability. Acetylcholine and norepinephrine should positively bias perception, attention, and cognition rather than raw affective amplification. Dopamine should enhance cognition and volition, but not so strongly that the system loses the well-known "inverted-U" restraint on prefrontal function. [7]

The leak matrix is diagonal and intentionally stronger than 1 on every channel so that $C_i = 1$ always points inward while $C_i = 0$ always points inward, making the state cube forward invariant without altering your ODE form:

$$\Lambda = \mathrm{diag}(1.30,\ 1.25,\ 1.20,\ 1.25,\ 1.35,\ 1.35,\ 1.15,\ 1.10,\ 1.15,\ 1.10,\ 1.05).$$

The three scalar-input gain matrices are diagonal, implemented from the following gain vectors:

$$w_E = \begin{bmatrix} 0.20 \\ 0.05 \\ 0.00 \\ 0.20 \\ 1.00 \\ 0.25 \\ 0.70 \\ 0.45 \\ 0.10 \\ 0.10 \\ -0.05 \end{bmatrix}, \qquad w_R = \begin{bmatrix} 0.20 \\ 1.00 \\ -0.15 \\ 0.05 \\ 0.25 \\ 0.10 \\ 0.00 \\ 0.30 \\ 0.55 \\ 0.35 \\ 0.10 \end{bmatrix}, \qquad w_N = \begin{bmatrix} 1.00 \\ 0.20 \\ -0.05 \\ 0.95 \\ 0.30 \\ 0.05 \\ 0.40 \\ 0.25 \\ 0.15 \\ 0.10 \\ -0.10 \end{bmatrix}.$$

Equivalently, $W_E = \mathrm{diag}(w_E)$, $W_R = \mathrm{diag}(w_R)$, and $W_N = \mathrm{diag}(w_N)$. These choices satisfy your stated hard constraints exactly: high RPE most strongly drives DA; high novelty most strongly drives ACh and NE; E_{err} most strongly drives Glu; inhibitory balancing is carried by $GABA$ and net-serotonergic suppression. [8]

The recurrent state-coupling matrix W_C is given exactly below. Rows are targets and columns are sources in the state order above.

```
W_C = np.array([
    [ 0.05,  0.00, -0.08,  0.08,  0.10, -0.12,  0.10,  0.06,  0.08,  0.06,
0.00],  # NE
    [ 0.06,  0.08, -0.12,  0.02,  0.05, -0.10,  0.00,  0.10,  0.18,  0.16,
0.08],  # DA
```

```
    [-0.04, -0.06,  0.10,  0.00, -0.08,  0.10, -0.02,  0.00, -0.06, -0.04,
0.12],  # 5HT
    [ 0.12,  0.00, -0.04,  0.08,  0.10, -0.08,  0.14,  0.12,  0.00,  0.04,
-0.02],  # ACh
    [ 0.06,  0.08, -0.10,  0.06,  0.10, -0.16,  0.12,  0.14,  0.08,  0.10,
0.00],  # Glu
    [-0.02, -0.02,  0.10, -0.02,  0.12,  0.06,  0.00,  0.00,  0.05, -0.02,
0.04],  # GABA
    [ 0.16,  0.02, -0.06,  0.18,  0.18, -0.12,  0.12,  0.12,  0.00,  0.00,
-0.04],  # Perception
    [ 0.10,  0.12,  0.00,  0.10,  0.14, -0.10,  0.18,  0.16,  0.05,  0.12,
0.08],  # Cognition
    [ 0.18,  0.10,  0.12,  0.00,  0.05, -0.18,  0.04,  0.04,  0.12,  0.12,
0.10],  # Affect
    [ 0.08,  0.18, -0.04,  0.02,  0.12, -0.10,  0.00,  0.16,  0.16,  0.16,
0.08],  # Volition
    [ 0.04,  0.08,  0.16,  0.00,  0.00, -0.10,  0.02,  0.08,  0.12,  0.14,
0.18],  # Self
], dtype=float)
```

These are **normalized architecture coefficients**, not claims about literal synaptic efficacies in vivo. The ratification claim is narrower and stronger: the signs and relative magnitudes follow validated neurocomputational directionality, while the absolute scale is deliberately chosen to satisfy the global contraction inequality of the bounded sigmoidal ODE. That is the proper level of rigor for an executable architecture matrix. [9]

Stability and bounded energy proof

For the 11D ODE,

$$\dot{C} = -\Lambda C + \sigma(W_C C + w_E E_{err} + w_R RPE + w_N Novelty + b),$$

the logistic nonlinearity is globally Lipschitz with constant $L_\sigma = \frac{1}{4}$. For two trajectories C_1, C_2, letting $\Delta = C_1 - C_2$ and $V = \frac{1}{2}\|\Delta\|_2^2$, one gets

$$\dot{V} \le -\left(\lambda_{\min}(\Lambda) - \frac{1}{4}\|W_C\|_2\right)\|\Delta\|_2^2.$$

A sufficient condition for global exponential contraction is therefore

$$\lambda_{\min}(\Lambda) > \frac{1}{4}\|W_C\|_2.$$

With the matrix above, $\lambda_{\min}(\Lambda) = 1.05$ and $\|W_C\|_2 \approx 0.867$, so the contraction margin is about $0.833 > 0$. This places the recurrent block comfortably inside the globally stable regime described for sigmoidal neural systems with Lipschitz activations and Lyapunov-diagonally-stable structure. [10]

Boundedness is even stronger than contraction here. Because $\sigma(\cdot) \in (0, 1)$, every coordinate satisfies

$$\dot{C}_i \geq 0 \quad \text{at } C_i = 0, \qquad \dot{C}_i \leq -\lambda_i + 1 < 0 \quad \text{at } C_i = 1,$$

since every $\lambda_i > 1$. Therefore the cube $[0,1]^{11}$ is forward invariant. That directly fixes the kind of out-of-range drift your empirical log warned about, without changing the documented ODE form.

For the resource law, boundedness follows from the boundary derivatives already noted. The energy guarantee is exact because

$$-\dot{B}(t) = \frac{p(t)}{0.85 E_{\max}} B(t) \geq 0$$

and hence

$$\int_0^T \left(-\dot{B}(t)\right)\, dt = B(0) - B(T) \leq 0.85 E_{\max}.$$

So the admitted bridge expenditure can never exceed the allowed 85% reservoir. That is the clean Lyapunov-compatible way to encode conservation while still permitting a slow recovery of **accessible** resources C_r toward the remaining budget ceiling. [11]

For the 666D term, using $\widehat{\alpha}(M)$ means that any coupling mode with positive real growth immediately produces a monotone increase in Π_M, and because the penalty is exponential in $[\widehat{\alpha}(M)]_+$, the gate becomes sharply more restrictive exactly when the coupling layer is mathematically closest to unstable amplification. That is the bridge equivalent of choking off active support before recurrent resonance can bleed into persistent drift or hallucination-like runaway attractors. [12]

NumPy reference implementation

The code below is a bare-metal NumPy reference that implements one timestep of the ratified formulas and returns the updated C_r and the finalized integer k_{allow}. It preserves your documented architecture: bounded 11D ODE, affine 11D sparsity gate, spectral 666D penalty, and the existing fail-safe trigger $\|\dot{C}\|_2 > 5C_r$. It does not alter the already-verified masking invariant on the discrete substrate; that masking step remains exactly where your empirical note placed it.

```
import numpy as np


def sigma(x: np.ndarray) -> np.ndarray:
    """Logistic activation."""
    return 1.0 / (1.0 + np.exp(-x))


# State order:
# [NE, DA, 5HT, ACh, Glu, GABA, Perception, Cognition, Affect, Volition, Self]

LAMBDA = np.diag([
```

```
    1.30, 1.25, 1.20, 1.25, 1.35, 1.35, 1.15, 1.10, 1.15, 1.10, 1.05
]).astype(float)

W_C = np.array([
    [ 0.05,  0.00, -0.08,  0.08,  0.10, -0.12,  0.10,  0.06,  0.08,  0.06,
0.00],
    [ 0.06,  0.08, -0.12,  0.02,  0.05, -0.10,  0.00,  0.10,  0.18,  0.16,
0.08],
    [-0.04, -0.06,  0.10,  0.00, -0.08,  0.10, -0.02,  0.00, -0.06, -0.04,
0.12],
    [ 0.12,  0.00, -0.04,  0.08,  0.10, -0.08,  0.14,  0.12,  0.00,  0.04,
-0.02],
    [ 0.06,  0.08, -0.10,  0.06,  0.10, -0.16,  0.12,  0.14,  0.08,  0.10,
0.00],
    [-0.02, -0.02,  0.10, -0.02,  0.12,  0.06,  0.00,  0.00,  0.05, -0.02,
0.04],
    [ 0.16,  0.02, -0.06,  0.18,  0.18, -0.12,  0.12,  0.12,  0.00,  0.00,
-0.04],
    [ 0.10,  0.12,  0.00,  0.10,  0.14, -0.10,  0.18,  0.16,  0.05,  0.12,
0.08],
    [ 0.18,  0.10,  0.12,  0.00,  0.05, -0.18,  0.04,  0.04,  0.12,  0.12,
0.10],
    [ 0.08,  0.18, -0.04,  0.02,  0.12, -0.10,  0.00,  0.16,  0.16,  0.16,
0.08],
    [ 0.04,  0.08,  0.16,  0.00,  0.00, -0.10,  0.02,  0.08,  0.12,  0.14,
0.18],
], dtype=float)

# Scalar-input gain vectors; equivalent to diagonal matrices W_E, W_R, W_N
W_E = np.array([0.20, 0.05, 0.00, 0.20, 1.00, 0.25, 0.70, 0.45, 0.10, 0.10,
-0.05], dtype=float)
W_R = np.array([0.20, 1.00, -0.15, 0.05, 0.25, 0.10, 0.00, 0.30, 0.55, 0.35,
0.10], dtype=float)
W_N = np.array([1.00, 0.20, -0.05, 0.95, 0.30, 0.05, 0.40, 0.25, 0.15, 0.10,
-0.10], dtype=float)

# Basal offset
BIAS = np.array([0.10, 0.08, 0.12, 0.10, 0.05, 0.10, 0.08, 0.10, 0.06, 0.06,
0.08], dtype=float)

# Resource-law gains
ALPHA_C = 0.25
ALPHA_R = 0.18
ALPHA_N = 0.14
RHO_CR = 0.08

# 11D sparsity gate coefficients; keeps the documented affine form
K0 = 256.0
```

```
A_DA = 96.0
B_5HT = 64.0
C_GABA = 72.0
D_TEMP = 32.0

# Spectral friction gains
GAMMA_M = 28.0
BETA_M = 2.4


def coupling_penalty(M: np.ndarray) -> tuple[float, float]:
    """
    Return (effective_resonance_score, scalar_penalty).

    Uses:
      alpha(M)  = max Re(lambda_i(M))
      mu2(M)    = lambda_max((M + M.T)/2)
      alpha_hat = max(alpha, mu2)

    The second term is a conservative non-normality guard.
    """
    if M.ndim != 2 or M.shape[0] != M.shape[1]:
        raise ValueError("M must be a square 2D NumPy array.")

    alpha = float(np.max(np.real(np.linalg.eigvals(M))))
    mu2 = float(np.max(np.linalg.eigvalsh((M + M.T) / 2.0)))
    alpha_hat = max(alpha, mu2)

    penalty = GAMMA_M * (np.exp(BETA_M * max(alpha_hat, 0.0)) - 1.0)
    return alpha_hat, float(penalty)


def bridge_timestep(
    C: np.ndarray,
    C_r: float,
    budget_B: float,
    M: np.ndarray,
    E_err: float,
    RPE: float,
    Novelty: float,
    dt: float = 0.02,
    E_max: float = 1.0,
) -> tuple[np.ndarray, float, float, int, dict]:
    """
    Execute one bridge timestep.

    Parameters
    ----------
```

```
    C : (11,) array
        Current continuous state in [0, 1]^11.
    C_r : float
        Cognitive resources in [0.05, 1.0].
    budget_B : float
        Remaining expendable bridge budget in [0, 0.85 * E_max].
    M : (666, 666) array
        Continuous coupling/intermodulation matrix.
    E_err, RPE, Novelty : float
        Scalar drives.
    dt : float
        Integration timestep.
    E_max : float
        Resource budget scale.

    Returns
    -------
    C_next : np.ndarray
    C_r_next : float
    budget_B_next : float
    k_allow : int
    diagnostics : dict
    """
    C = np.asarray(C, dtype=float).reshape(-1)
    if C.shape[0] != 11:
        raise ValueError("C must have shape (11,).")
    if not (0.05 <= C_r <= 1.0):
        raise ValueError("C_r must be in [0.05, 1.0].")
    if budget_B < 0.0 or budget_B > 0.85 * E_max:
        raise ValueError("budget_B must be in [0, 0.85 * E_max].")

    # Continuous ODE
    drive = W_C @ C + W_E * E_err + W_R * RPE + W_N * Novelty + BIAS
    dCdt = -LAMBDA @ C + sigma(drive)
    C_next = np.clip(C + dt * dCdt, 0.0, 1.0)

    # Energy / resource dynamics
    p = (
        ALPHA_C * float(np.dot(dCdt, dCdt))
        + ALPHA_R * float(RPE ** 2)
        + ALPHA_N * float(Novelty ** 2)
    )

    # Exact 85% budget law
    dBdt = -(p / (0.85 * E_max)) * budget_B
    budget_B_next = float(np.clip(budget_B + dt * dBdt, 0.0, 0.85 * E_max))

    # Cognitive resource law
```

```
    budget_ceiling = 0.05 + 0.95 * (budget_B / (0.85 * E_max))
    dCrdt = RHO_CR * (budget_ceiling - C_r) - p * (C_r - 0.05)
    C_r_next = float(np.clip(C_r + dt * dCrdt, 0.05, 1.0))

    # 11D gate
    DA = float(C_next[1])
    HT5 = float(C_next[2])
    GABA = float(C_next[5])
    Temp = 1.0 - float(C_next[8])  # preserved from the existing bridge formula

    k_11D = K0 + A_DA * DA - B_5HT * HT5 - C_GABA * GABA - D_TEMP * Temp

    # 666D spectral friction
    alpha_hat, penalty_M = coupling_penalty(M)

    # Final threshold before fail-safe
    k_allow = int(np.clip(np.floor(k_11D - penalty_M), 1, 307))

    # Documented Lyapunov fail-safe
    norm_dCdt = float(np.linalg.norm(dCdt))
    unstable = norm_dCdt > (5.0 * C_r_next)
    if unstable:
        k_allow = max(1, int(np.floor(0.5 * k_allow)))

    diagnostics = {
        "drive": drive,
        "dCdt": dCdt,
        "power_p": p,
        "budget_ceiling": budget_ceiling,
        "k_11D": float(k_11D),
        "alpha_hat": alpha_hat,
        "penalty_M": penalty_M,
        "norm_dCdt": norm_dCdt,
        "unstable": unstable,
    }
    return C_next, C_r_next, budget_B_next, k_allow, diagnostics
```

This implementation is the minimal ratified bridge I would sign off on. It respects the already-verified discrete masking invariant from your July 18 log, it gives you an exact dC_r/dt with an exact 85% energy cap, it converts the 666D layer into a real spectral friction law rather than an RMS placeholder, and it instantiates the 11D ODE couplings with explicit normalized weights that are simultaneously biologically legible and Lyapunov-safe. filecite turn0file0

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1 7 8 A neural substrate of prediction and reward - PubMed
https://pubmed.ncbi.nlm.nih.gov/9054347/?utm_source=chatgpt.com

2 3 4 11 An energy budget for signaling in the grey matter of the brain - PubMed
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5 6 12 Necessary and sufficient condition for absolute stability of normal neural networks - PubMed
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9 Quantifying the inverted U: A meta-analysis of prefrontal dopamine, D1 receptors, and working memory - PubMed
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